

+ Code

+ Text

[] car_price_prediction.ipynb

[] car_data.csv

[1] 1s
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

[19] 0s
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 301 entries, 0 to 300
Data columns (total 9 columns):
Column Non-Null Count Dtype

0 Car_Name 301 non-null object
1 Year 301 non-null int64

qt.describe()

	Year	Selling_Price	Present_Price	Driven_kms	Owner
count	301.000000	301.000000	301.000000	301.000000	301.000000
mean	2013.627907	4.661296	7.628472	36947.205980	0.043189
std	2.891554	5.082812	8.642584	38886.883882	0.247915
min	2003.000000	0.100000	0.320000	500.000000	0.000000
25%	2012.000000	0.900000	1.200000	15000.000000	0.000000
50%	2014.000000	3.600000	6.400000	32000.000000	0.000000
75%	2016.000000	6.000000	9.900000	48767.000000	0.000000
max	2018.000000	35.000000	92.600000	500000.000000	3.000000

[24] print(df.columns)

Index(['Car_Name', 'Year', 'Selling_Price', 'Present_Price', 'Driven_kms', 'Fuel_Type', 'Selling_type', 'Transmission', 'Owner'],

```
[27] df.drop('Car_Name', axis=1, inplace=True)
```

```
[28] df = pd.get_dummies(df, drop_first=True)
```

```
[29] df.head()
```

	Year	Selling_Price	Present_Price	Driven_kms	Owner	Car_Age	Fuel_Type_Diesel	Fuel_Type_Petrol	Selling_type_Individual	Transmission_M
0	2014	3.35	5.59	27000	0	12	False	True	False	
1	2013	4.75	9.54	43000	0	13	True	False	False	
2	2017	7.25	9.85	6900	0	9	False	True	False	
3	2011	2.85	4.15	5200	0	15	False	True	False	
4	2014	4.60	6.87	42450	0	12	True	False	False	

```
[20]
✓ 0s df.isnull().sum()

...
0
Car_Name 0
Year 0
Selling_Price 0
Present_Price 0
Driven_kms 0
Fuel_Type 0
Selling_type 0
Transmission 0
Owner 0

dtype: int64
```

```
[21] df.dropna(inplace=True)
[22] df.isnull().sum()
```

...	0
Car_Name	0
Year	0
Selling_Price	0
Present_Price	0
Driven_kms	0
Fuel_Type	0
Selling_type	0
Transmission	0
Owner	0


```
[30] X = df.drop(
      'Selling_Price',
      axis=1
    )

    y = df['Selling_Price']

[31] from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

[32] from sklearn.linear_model import LinearRegression

model = LinearRegression()

model.fit()
```

+ Code + Text

```
[32] ✓ 0s
from sklearn.linear_model import LinearRegression

model = LinearRegression()

model.fit(
    X_train,
    y_train
)
```

LinearRegression
LinearRegression()

```
[33] ✓ 0s
predictions = model.predict(X_test)

print(predictions[:10])

[ 2.95433731  8.17716341  6.45612271 -1.42337164  9.08864657  7.41793553
 1.33513921  0.84032259  1.36320242  7.49067757]
```

```
[34] ✓ 0s
from sklearn.metrics import mean_absolute_error

mae = mean_absolute_error(
    y_test,
    predictions
)

print("MAE:", mae)

MAE: 1.216374019332973
```

```
[35] ✓ 0s
from sklearn.metrics import r2_score

r2 = r2_score(
    y_test,
    predictions
)

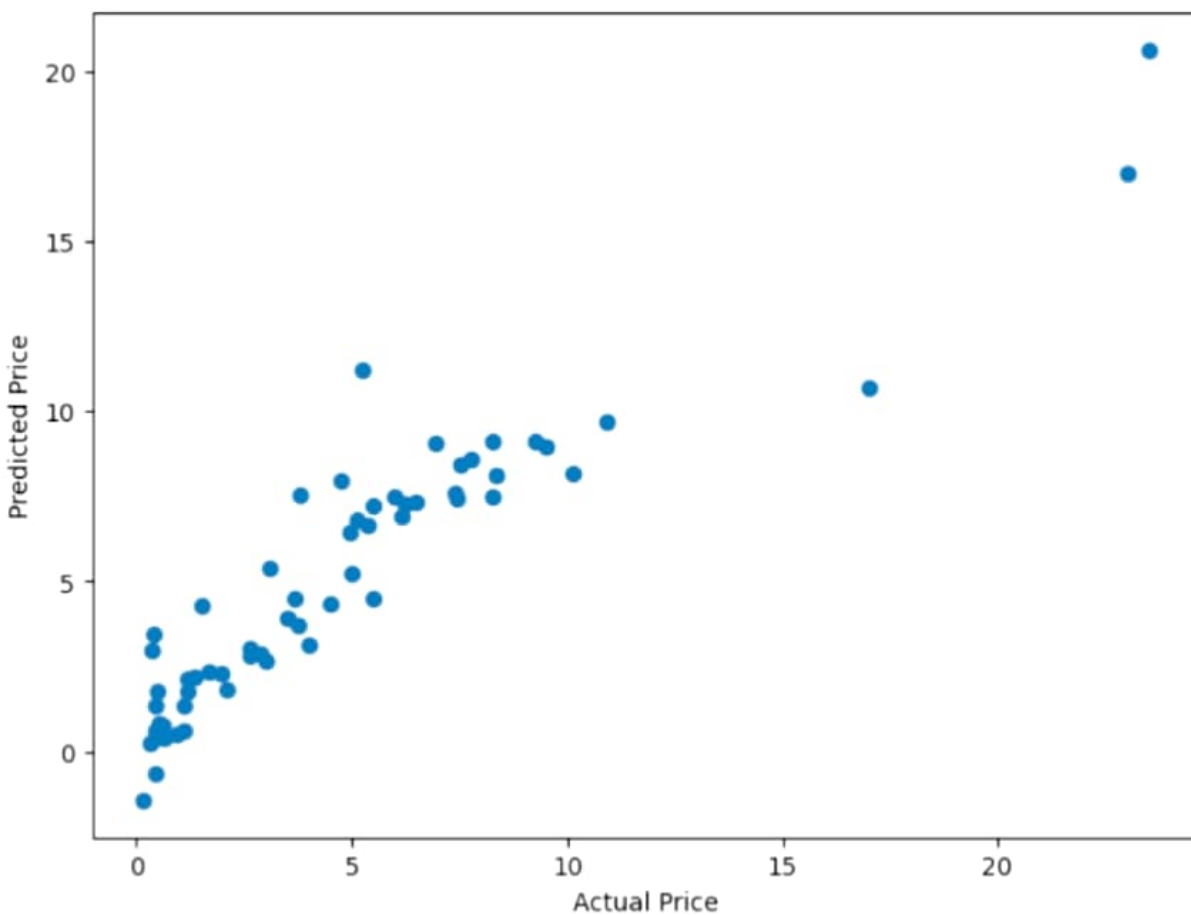
print("R2 Score:", r2)
```


⋮
R2 Score: 0.8488707839192597

```
[36] ▶ plt.figure(figsize=(8,6))  
✓ 1s  
  
plt.scatter(  
    y_test,  
    predictions  
)  
  
plt.xlabel("Actual Price")  
plt.ylabel("Predicted Price")  
  
plt.title(  
    "Actual vs Predicted Car Prices"  
)  
  
plt.show()
```



Actual vs Predicted Car Prices



```
[37] ✓ 0s
coefficients = pd.DataFrame({
    'Feature': X.columns,
    'Coefficient': model.coef_
})

print(coefficients)
```

	Feature	Coefficient
0	Year	0.176463
1	Present_Price	0.429070
2	Driven_kms	-0.000006
3	Owner	-0.905018
4	Car_Age	-0.176463
5	Fuel_Type_Diesel	2.534904
6	Fuel_Type_Petrol	0.739155
7	Selling_type_Individual	-1.191642
8	Transmission_Manual	-1.640074

```
[38] ✓ 0s
sample = X.iloc[[0]]
```

```
6      Fuel_Type_Petrol      0.739155
7  Selling_type_Individual  -1.191642
8      Transmission_Manual  -1.640074
```

```
[38] ✓ 0s
sample = X.iloc[[0]]

predicted_price = model.predict(sample)

print(predicted_price)
```

```
[3.93544304]
```

```
[39] ✓ 0s
import pickle

pickle.dump(
    model,
    open(
        "car_price_model.pkl",
        "wb"
    )
)
```

```
[3.93544304]  
  
[39]  
✓ 0s  
import pickle  
  
pickle.dump(  
    model,  
    open(  
        "car_price_model.pkl",  
        "wb"  
    )  
)
```